Navier-Stokes Equations, Boundary Layer Separation & Drag

Differential Navier-Stokes derivations, Prandtl boundary layer equations, momentum thickness, flow separation control, and lift-to-drag curves.

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1. Navier-Stokes Momentum Equations

Balances inertial forces against pressure gradients and viscous diffusion: rho*(DV/Dt) = -grad(p) + mu*nabla^2(V) + rho*g.

2. Prandtl Boundary Layer Theory

Viscous deceleration is confined to a thin layer delta near solid boundaries. Adverse pressure gradients trigger boundary layer separation.